

**Q5: A surveillance system requires real-time image processing. Classify the level of Computer Vision involved and justify your choice.**

### Introduction

Computer Vision is a branch of Artificial Intelligence that enables computers to acquire, process, analyze, and interpret images or videos. Depending on the complexity of the task, computer vision can be classified into different levels. A surveillance system that processes live video requires fast analysis and decision-making, making it an example of High-Level Computer Vision.

### Classification of the Computer Vision Level

A real-time surveillance system is classified as High-Level Computer Vision.

### Reason

High-level computer vision not only processes images but also understands their content and makes intelligent decisions. A surveillance system continuously captures live video, detects moving objects, recognizes people or vehicles, tracks their movement, and identifies suspicious activities automatically.

Unlike low-level image processing, which only improves image quality, high-level computer vision extracts meaningful information from the scene and supports automated monitoring.

### Justification

A surveillance system performs the following high-level computer vision tasks:

#### 1. Image Acquisition

- Captures live video continuously using surveillance cameras.
- Each video frame is treated as a digital image for processing.

#### 2. Object Detection

- Detects humans, vehicles, animals, or other moving objects in the scene.

#### 3. Object Recognition

- Identifies the detected object, such as distinguishing a person from a vehicle.

#### 4. Object Tracking

- Tracks the movement of detected objects across multiple video frames.

#### 5. Activity Analysis

- Analyzes the behavior of objects.
- Detects abnormal or suspicious activities such as intrusion, theft, or unauthorized access.

#### 6. Decision Making

- Generates alerts or notifications automatically when unusual events are detected.
- This requires understanding the scene rather than simply enhancing the image.

#### Applications

- Airport security
- Railway station monitoring
- Traffic surveillance
- Smart city monitoring
- Bank and ATM security
- Industrial safety monitoring

#### Conclusion

A real-time surveillance system belongs to the High-Level Computer Vision category because it not only captures and processes images but also detects, recognizes, tracks, and analyzes objects to make intelligent decisions automatically. These capabilities are essential for efficient and reliable security monitoring.